

MovieLOD

DBpedia reuse · OWL reasoning · linked movie knowledge

From movie data to a knowledge graph

Ontology · RDF · Linked Open Data · SPARQL

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30

real films

37

named classes

1,727

external links

Objectives and synchronized evidence

Ontology, data, publication, linking and SPARQL access

ID	Requirement	Deliverable / evidence
R1	DBpedia-based ontology	37 named classes; HermiT consistency
R2	Source-supported data	76 retained responses and provenance
R3	RDF publication	Canonical RDF / OWL 3.0.0
R4	Identity alignment	1,699 Wikidata + 28 DBpedia links
R5	SPARQL access	27 questions; source / inferred / named graph modes

Report: 15 pages · main slides: 24 · MP4 excluded by request. Examples: Inception and Nolan.

Architecture and knowledge layers

One reproducible pipeline across every interface

01

SOURCE FACTS

76 retained responses
+ provenance

02

BUILD 3.0.0

Canonical RDF / OWL
+ resource pages

03

REASON / CHECK

HermiT + OWL RL
27 query checks

04

QUERY MODES

Source / inference
Before / after

NEW MODEL: asserted.ttl + schema.ttl + inferred.ttl; query 27 uses before_after.trig.
APPLICATION: Flask / RDFLib / Comunica share the canonical version-3 graph exports.

Dataset: units and counting scope

Final OWL: 19,025 triples including schema and asserted facts

30

films

851

people

1,010

credit records

45

companies

672

award entities

76

source responses

37 named classes · 19 object + 5 datatype properties · 4 controlled role individuals

Ontology inventory: reuse before extension

17 DBpedia classes + 1 VoID class + 19 local classes

17 classes	DBpedia classes and hierarchy
1 class	void:Dataset
3 classes	Contribution / Role / SourceSnapshot
2 classes	ActionGenre / DramaGenre
4 classes	Role-defined credit subclasses
10 classes	Domain inferred subsets

37 named classes: 17 dbo: + 1 void: + 19 ex:
10 inferred subsets + 4 supporting credit types.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P00

IRI và phiên bản 3.0.0

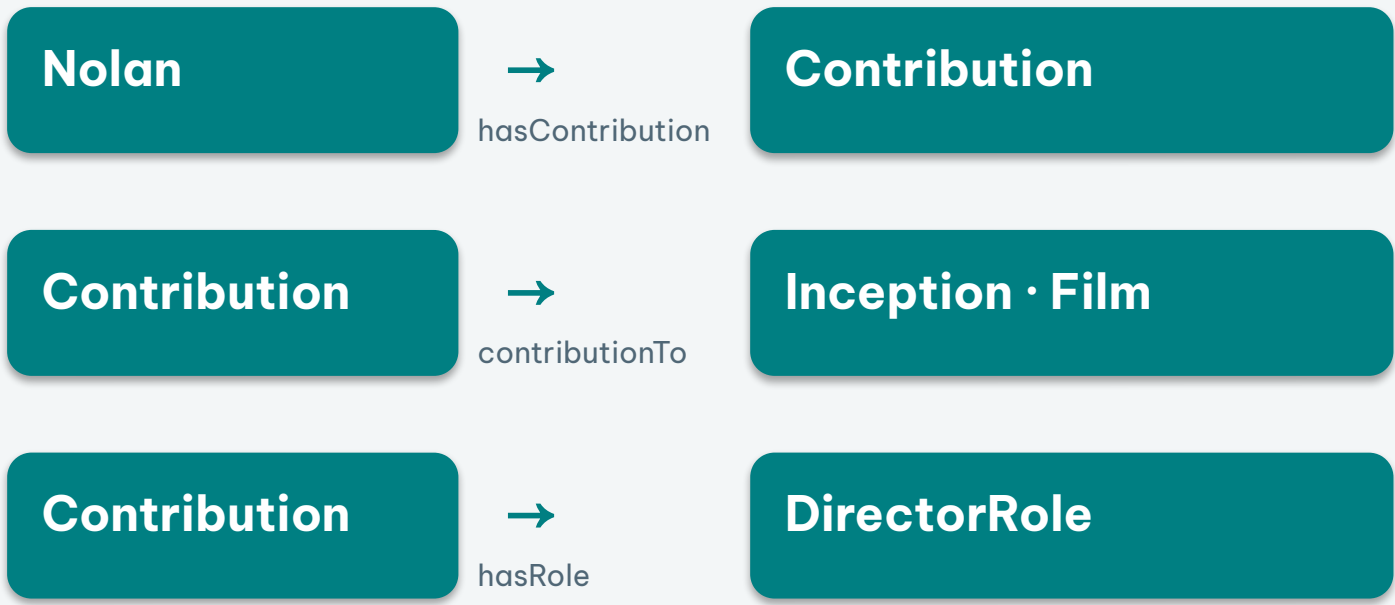
P00_ontology_header.png

1. Mở ontology/Movie_Knowledge_Graph.owl.
2. Chọn Active Ontology / Ontology Header.
3. Giữ IRI và versionInfo 3.0.0 trong ảnh.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Contribution: person, film and role

Exactly one endpoint of each kind per credit record



Each role has its own record

Director / Actor / Writer / Producer. Nolan has 3 roles in Inception.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P04

Ràng buộc Contribution

P04_contribution_restrictions.png

- 1. Chọn ex:Contribution.
- 2. Hiện exactly 1 cho contributionBy, contributionTo, hasRole.
- 3. Chọn hasRole để thấy Functional và role individual.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Properties: reuse, inverse and chain

19 object properties; custom relations describe credit context

Property	Domain → range	Inverse
contributionBy	Credit → Person	hasContribution
contributionTo	Credit → Film	contributionOf
hasRole	Credit → Role	functional
dbo:director	Film → Person	ex:directed
dbo:starring	Work → Actor	ex:actedIn
contributedTo	Person → Film	property chain

ẢNH PROTÉGÉ CẦN BỔ SUNG · P05

Domain / range / inverse

P05_object_property.png

- 1. Chọn contributionBy: Contribution → dbo:Person.
- 2. Hiện Functional và inverse hasContribution.
- 3. Có thể chụp contributedTo với property chain.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

contributionBy / contributionTo / hasRole: functional. Domain/range can infer types; inverse reverses a relation.

Verified domain classes after HermiT

All listed types have zero assertions in the input data

Class	Condition in plain language	Count	Method
Filmmaker	Directing, writing or producing credit	89	HermiT
ActionFilm	Action genre membership	12	HermiT
AwardWinningFilm	Film with an award	26	HermiT
MultiCreditContributor	At least 2 provably distinct credits	17	HermiT
ThreeCreditContributor	At least 3 provably distinct credits	7	HermiT
WriterDirector	MovieDirector AND ScreenWriter	10	HermiT
ActorFilmmaker	Actor AND Filmmaker	7	HermiT
AwardWinningFilmmaker	Filmmaker with an award	65	HermiT
AwardWinningActionFilm	ActionFilm AND AwardWinningFilm	10	HermiT
GenreCrossingFilm	Action AND drama genre membership	8	HermiT

Supporting credit types: Acting 855 · Directing 31 · Writing 51 · Producing 73; dbo:Actor 769.

Nolan: facts → credit type → Filmmaker

A multi-step explanation grounded in real records

SOURCE FACTS

Nolan a dbo:Person; credit hasRole DirectorRole.



CREDIT CLASS

credit a DirectingContribution; inverse yields
hasContribution.



PERSON CLASS

Nolan a Filmmaker; directing + writing also yield
WriterDirector.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P06

Định nghĩa Filmmaker

P06_filmmaker_equivalent_class.png

1. Chọn ex:Filmmaker.
2. Hiện Equivalent To với Person AND các nhánh SOME.
3. Đọc các nhánh Directing / Writing / ProducingContribution.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Nolan: 8 directed films, 22 credit IRIs in the sample; three roles on Inception.

Genuine minimum-cardinality reasoning

Functional hasRole + different roles prove distinct credits

MultiCreditContributor · 17

Person and hasContribution min 2 Contribution.
Distinctness needs semantic evidence.

ThreeCreditContributor · 7

Person and hasContribution min 3 Contribution.
Nolan supplies three different-role witnesses.

Distinct IRIs ≠ provably different OWL individuals.

Negative test: min 2 genres gives 0 DL members; SQL/SPARQL counting is a different claim.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P07

Cardinality trên Contribution

P07_three_credit_cardinality.png

1. Chọn ex:ThreeCreditContributor; hiện min 3 hasContribution Contribution.
2. Chạy HermiT; chọn Nolan ở inferred view.
3. Chụp thêm Functional hasRole và AllDifferent của bốn role.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Linked data: identity and publication

Canonical identifiers, graph exports and external identity links

1★	Open license	CC BY-SA 4.0
2★	Structured source descriptions	Claims, identifiers, values
3★	Open RDF serializations	RDF/XML, Turtle, JSON-LD
4★	HTTP identifiers and RDF	Canonical version 3.0.0
5★	External identity links	1,699 Wikidata + 28 DBpedia

Public graphs and local canonical exports are checked for semantic equality.

Verification of the final OWL

Direct HermiT run; evidence bound to the file hash

PASS

HermiT consistency

0

unsatisfiable named classes

27

design queries executed

965

chain pairs verified

Final OWL: HermiT verification

Direct RDF/XML input · version 3.0.0 · counts restricted to local IRIs

Consistency	PASS
Unsatisfiable named classes	0
Design SPARQL queries	27
Property-chain person-film pairs	965
MultiCredit / ThreeCredit	17 / 7
WriterDirector	10
Web publication scope	Canonical 3.0.0

Author-rendered evidence summary; not a screenshot of Protégé.

Canonical OWL verified by HermiT; browser queries use matching source, reasoned and named graphs.

Evidence capture and submission checklist

Report: 15 pages · main deck: 24 slides · no MP4 included

Image	Slide	Evidence to capture
P00	5	Ontology 3.0.0 header
P04	6	Contribution restrictions
P05	7	Inverse and functional properties
P06	9	Filmmaker equivalent class
P07	10	Three-credit cardinality

Before submission

1. Verify ontology version 3.0.0.
2. Insert the requested genuine screenshots.
3. Save them in evidence/protege/ for automatic replacement.
4. Rehearse the script and demonstrate the current query modes.

Use the canonical OWL for Protégé evidence; source facts and inferred knowledge have separate scopes.